Design, Sensing, and Control of a Novel UAV Platform for Aerial Drilling and Screwing

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***Abstract*—Hole drilling and bolt screwing are frequently performed tasks in construction, decoration, and maintenance. Traditionally, sending human workers to perform these tasks in hard-to-reach locations is both dangerous and costly. In this paper, we present an aerial manipulation platform that allows a human user to remotely conduct omnidirectional drilling and screwing. The design of the platform features a quadrotor UAV with each pair of rotors independently tilted by a servo, forming an "H" configuration, on which a 1-DOF manipulator carrying a motorized drill or screw driver is mounted. With such a design, the end-effector can face any direction on the longitudinal plane and exert a big enough contact force for drilling and screwing without the need of changing the vehicle body's orientation. Compared to previous UAVs that can only drill holes vertically into the ground, the proposed design also allows horizonal drilling/screwing into a wall or a cliff, making it suitable for a vast range of real applications. Based on the dynamic equations of the system, a dual-level control law is proposed. The low-level attitude controller uses an adaptive robust control (ARC) to accurately regulate the attitude angles in the presence of force/torque uncertainties that may occur during the drilling and screwing process, while a selective impedance controller is implemented at high level to indirectly control the contact force commanded by the user. In addition, a vision-based real-time target identification and tracking method integrating a YOLO v3 real-time object detector with feature tracking, and morphological operations is developed to identify and track the target point for drilling and screwing specified by the user. Various in-lab experiments on a self-made prototype demonstrate the feasibility and effectiveness of the proposed approach for aerial drilling and screwing.**

***Index Terms*—Aerial manipulation, drilling, screwing, UAV design, impedance control, power efficiency, vision tracking.**

I. Introduction

As Unmanned Aerial Vehicles (UAVs) are technologically maturing, novel applications of UAVs have sprung forth that exploit and extend their inherent aerial capabilities. In particular, an increasing number of applications require UAVs to actively interact with environment rather than simply serving as a passive sensing platform. Among them, aerial manipulation tasks, such as grasping [1], fetching [2], writing [3], peg-in-hole [4], and object trasporting are of particular interest to researchers, as they can potentially replace human workers in a variety of construction, maintenance, and transportation tasks to be performed at hard-to-reach or dangerous locations. However, these tasks are at an entry level and require only very basic interaction between UAV and environment. Many practically meaningful applications that involve sophisticated contact and force control remain hardly achievable under the traditional design and control of aerial manipulators. Particularly, hole drilling and screwing tasks, as two of the most common and essential operations to be conducted together in engineering, have only been touched in the aerial manipulation scenario to a very limited extent, e.g., vertical drilling down to the ground or a workpiece [5]–[8].

The frequently performed drilling and screwing operations share some features in common. Specifically, there are two forces (torques) present in both operations. One is the pushing (pressing) force normal to the surface of the workpiece, and the other is the torque around the normal axis. To achieve a successful drilling/screwing operation, a strong pushing force must be exerted, while the other degrees of freedom of the end-effector should be accurately position-controlled despite the reaction normal force and torque. Such a feature makes it problematic for traditional quadcopter platforms to implement drilling and screwing on a vertical wall no matter how the onboard manipulator is designed. This is because traditional quadcopters are highly underactuated, with the translational motion on the horizontal plane tightly coupled with rotational motion. In order to compensate for the strong contact force along the horizontal drilling/screwing direction, the body of the quadcopter has to rotate, letting the propellers face sideways to generate sufficient propulsion force. However, frequently tilting UAV's body to match the desired contact force will make the system highly unstable. Therefore, accurate position-level control along other degrees of freedom is impossible.

Thus, it is clear that the translation and rotation of the UAV body need to be independently controlled at least along a certain horizontal direction for a successful drilling/screwing. This requires the UAV body to have a degree of freedom of actuation higher than 4. Various fully-actuated (6-DOF) UAVs and 5-DOF actuated UAVs have been proposed up to date. Depending on the designs, they can be classified into two categories. The first one uses multiple fixed rotors facing different directions to achieve higher-DOF actuation [4], [9]–[11]. This type of design is structurally simple. However, since the rotors point to different directions, ineffective internal force cancellation will inevitably occur, lowering the power

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efficiency and load capacity of the vehicles. The second type of design employs a tilting-rotor structure, i.e., the introduction of one/two extra servo motors for each rotor to empower it with one/two-axis tilting ability [12]–[15]. With the additional tilting capability of the rotor axes, this type of design can partially or completely avoid ineffective force cancellation. However, the inclusion of a large number of movable servo parts makes the designs mechanically more complicated, with other potential problems such as higher weight and backlash from the intricate transmission system.

Besides the issue of the architecture and design, there still exist other challenges unique to the aerial drilling and screwing. For example, how to identify and precisely track the target for drilling/screwing, how to smoothly incorporate human users' command into the system, and how to achieve contact force regulation while maintaining accurate position-level control in other DOFs, remain to be solved. In this paper, we present a human-guided semi-automated aerial drilling/screwing platform based on a novel tilting-rotor aerial manipulator design. The design features a novel quadrotor UAV with each pair of rotors independently tilted by a servo, forming an "H" configuration, on which a simple 1-DOF manipulator carrying a motorized drill or screw driver is mounted, moving on the longitudinal plane. With only three additional servos needed (two for UAV body and one for manipulator) compared to the traditional coplanar quadcopters, the controls of the body's translational and rotational motions on the longitudinal plane are completely decoupled, and the end-effector can face any direction to apply a strong contact force for drilling and screwing on the longitudinal plane without the need of changing the vehicle body's orientation. Thus, the proposed platform can achieve omnidirectional drilling/screwing. Based on the dynamics of the proposed UAV design, a dual-level control law is proposed. The low-level attitude controller uses an adaptive robust control (ARC) to accurately regulate the attitude angles in the presence of force/torque uncertainties that may occur during the drilling and screwing process, while a selective impedance controller is implemented at high level to indirectly control the contact force commanded by the user. In addition, a vision-based guidance scheme is developed to identify and track the target feature on the workpiece. For verification, we conduct various indoor hole drilling and bolt screwing experiments on a vertical wood plate to show the applicability of our approach.

II. System Architecture And Mechanical Design

From the previous discussion, it is clear that the key to successfully implementing ominidirectional aerial drilling/screwing is to allow the end-effector face any desired feeding direction and exert a sufficiently large force while maintaining a stable UAV posture. To solve this challenge, we develop a novel quadrotor with two tiltable arms as the UAV body, on which a 1-DOF manipulator carrying a motorized drill or screw driver is mounted, as shown in Fig. 1. Specifically, the UAV platform consists of two pairs of rotors mounted on two independently-actuated arms placed at both sides of the vehicle in an "H" configuration, similar to [16], [17]. Each arm is tilted by a single servomotor, allowing independent adjustment of the thrust force direction of the pair of rotors it carries. Then, a 1-DOF lightweight manipulator carrying a self-designed drilling/screwing end-effector is mounted. The joint of the manipulator rotates on the longitudinal plane and is driven by a servo to align the drill/screwdriver with the desired feeding direction. To generate the spinning motion for the drilling/screwing process, we use a micro DC motor with $5V/2.5A$ power input to carry the tool and fix it to the end-effector. Additionally, a vision camera is installed on the manipulator. This eye-in-hand configuration is used to position and track a target point on an object of interest to complete aerial hole drilling and screwing tasks.

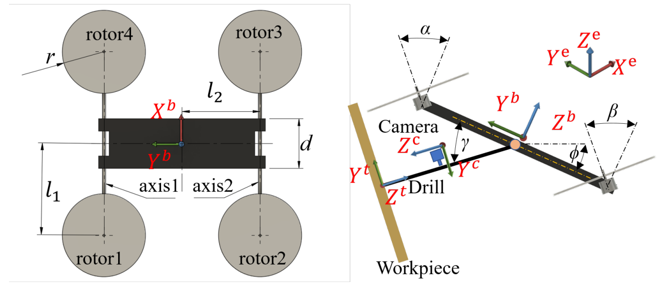


Fig. 1. The top view and front view of the proposed aerial drilling/screwing platform. α and β are the tilting angles of the two left motors and two right motors about axis1 and axis2 respectively. ϕ is the roll angle of the vehicle's body. γ is the rotating angle of the 1-DOF onboard manipulator.

Such a configuration has the following advantages which make it a perfect choice for aerial drilling/screwing platform:

1. By changing the two tilting angles α and β and the manipulator angle γ, the vehicle body can be actuated in 5 DOFs independently, while the end-effector can face any direction on the longitudinal plane ($Y^b Z^b$ plane) and independently exert a large enough contact force. This feature is particularly useful when conducting drilling/screwing on a vertical wall.
2. Ineffective internal propeller force cancellation can be avoided by setting $\alpha = \beta$, making the design energy efficient.
3. Only three additional servos (two for UAV body and one for manipulator) are used compared to traditional quadcopters with no extra transmission mechanism needed, leading to a less total weight and less probability of fault occurrence than other designs with too many servos.

III. System Modeling

A. Coordinate systems and assumptions

The following right-hand coordinate systems will be used throughout the rest of the paper, as shown in Fig. 1.

$\{e\}$: The earth (inertial) frame $\{O_e - X^e Y^e Z^e\}$.
$\{b\}$: The UAV body-fixed frame $\{O_b - X^b Y^b Z^b\}$.
$\{c\}$: The camera frame $\{O_c - X^c Y^c Z^c\}$.
$\{t\}$: The target frame $\{O_t - X^t Y^t Z^t\}$.

It is assumed that the origin of the body frame O_b is located at the center of mass of the UAV. The origin of the target frame O_t is located at the target point on the workpiece to drill or screw in, and the axis Z^t is perpendicular to the workpiece surface. Furthermore, since the 1-DOF onboard manipulator weights much less compared to the UAV body and its servo-motor has very quick response and high tracking accuracy, we can safely ignore the dynamics of the manipulator and simply set the manipulator angle $\gamma = \phi + \angle(Z^t, X^eY^e)$, where ϕ is the roll angle of the UAV body, and $\angle(Z^t, X^eY^e)$ is the angle between the drill/screw direction Z^t and the X^eY^e plane of the inertia frame. This choice of γ allows the end-effector to be aligned with the drilling/screwing direction.

B. Forces and Torques

Since the dynamics of the manipulator is ignored, all the contact forces and torques during the drilling/screwing process are directly transmitted to the UAV body. The total force acting on the UAV expressed in the body coordinate system is described as :

$$F^b = \begin{bmatrix} F_x^b & F_y^b & F_z^b \end{bmatrix}^T = F_p^b + F_g^b + F_c^b, \tag{1}$$

where $F_p^b = \sum_{i=1}^{4} F_{p_i}^b$ is the total thrust force which equals to the sum of all individual thrust forces expressed in body frame $\{b\}$. $F_g^b = mR_b^{e\,T} G$ is the gravitational force expressed in the body frame, m is the total mass of the UAV, $G = [0\ \ 0\ -g]^T$ is the gravitational acceleration vector in inertia frame. $F_c^b \in R^3$ is the lumped contact force between end-effector tool and the workpiece represented in body frame.

It is noted that since the propeller cannot rotate around Y^b, the projection of the thrust force onto X^b is equal to zero, i.e., $F_p^b = [0\ \ F_{py}^b\ \ F_{pz}^b]^T$. Defining F_1, F_2, F_3, F_4 as the absolute values of the thrust forces of the four propellers and α, β as the front and rear tilting angles, as shown in Fig. 1, the F_{py}^b and F_{pz}^b can be expressed as:

$$\begin{aligned} F_{py}^b &= (F_1 + F_4)S_\alpha + (F_2 + F_3)S_\beta, \\ F_{pz}^b &= (F_1 + F_4)C_\alpha + (F_2 + F_3)C_\beta. \end{aligned} \tag{2}$$

The total torque acting on the UAV expressed in the body frame is

$$\tau^b = \begin{bmatrix} \tau_x^b & \tau_y^b & \tau_z^b \end{bmatrix}^T = \tau_p^b + \tau_\Delta^b, \tag{3}$$

where $\tau_p^b = \sum_{i=1}^{4} r_i^b \times F_{p_i}^b + \sum_{i=1}^{4} D_i k F_{p_i}^b$ is the total torque generated by the propellers, including the thrust torque $\sum_{i=1}^{4} r_i^b \times F_{p_i}^b$ and drag torque $\sum_{i=1}^{4} D_i k F_{p_i}^b$, in which r_i^b is the coordinate vector of the center of the $i\text{-}th$ propeller in body frame, k is a constant ratio of drag torque to thrust force, and D_i is a constant equal to 1 when the $i\text{-}th$ propeller rotates in clockwise direction, and -1 when it rotates in counterclockwise direction. $\tau_\Delta^b \in R^3$ is the torque uncertainty incorporating the uncertain reaction torque between end-effector tool and the workpiece, and the torque caused by the contact force.

Defining l_1 as the half distance between the two rotors on each tilting axis, l_2 as the half distance between the two tilting axes, the x, y, z components of τ_p^b can be expressed as

$$\begin{aligned} \tau_{px}^b &= (F_1 + F_4)l_2C_\alpha - (F_2 + F_3)l_2C_\beta, \\ \tau_{py}^b &= (F_1 - F_4)(C_\alpha l_1 + S_\alpha k) + (F_2 - F_3)(C_\beta l_1 - S_\beta k), \\ \tau_{pz}^b &= (F_4 - F_1)(S_\alpha l_1 - C_\alpha k) + (F_3 - F_2)(S_\beta l_1 + C_\beta k). \end{aligned} \tag{4}$$

C. Dynamic Model

The system dynamics are derived in the Newton-Euler form as

$$\begin{bmatrix} F^b \\ \tau^b \end{bmatrix} = \begin{bmatrix} M & 0 \\ 0 & I^b \end{bmatrix} \begin{bmatrix} \dot{v}^b \\ \dot{\omega}^b \end{bmatrix} + \begin{bmatrix} \omega^b \times M v^b \\ \omega^b \times I^b \omega^b \end{bmatrix}, \tag{5}$$

where $v^b = \begin{bmatrix} v_x^b & v_y^b & v_z^b \end{bmatrix}^T$ and $\omega^b = \begin{bmatrix} \omega_x^b & \omega_y^b & \omega_z^b \end{bmatrix}^T$ are the vectors of vehicle's translational and angular velocities represented in body frame. I^b is the moment of inertia about the center of mass. $M = m \cdot I_3$ is the mass matrix.

D. Kinematic Relationship

Let $\lambda = \begin{bmatrix} x & y & z \end{bmatrix}^T$ be the vehicle's position (the coordinate of the body frame origin O_b) expressed in inertia frame $\{e\}$, $\eta = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T$ be the attitude vector of the vehicle. The kinematic relationship between the position/attitude of the vehicle and the body frame velocity is given by the following two equations:

$$\dot{\lambda} = R_b^e v^b, \quad \dot{\eta} = \Psi \omega^b, \tag{6}$$

where $\Psi = \begin{bmatrix} 1 & S_\phi T_\theta & C_\phi T_\theta \\ 0 & C_\phi & -S_\phi \\ 0 & \frac{S_\phi}{C_\theta} & \frac{C_\phi}{C_\theta} \end{bmatrix}$ is the transformation matrix at kinematic level.

IV. Control Design

Based on the dynamics, a dual-level control architecture is proposed as illustrated in Fig. 2. The low-level attitude controller uses an adaptive robust control (ARC) to accurately regulate the attitude angles in the presence of force/torque uncertainties that may occur during the drilling and screwing process, while a selective impedance controller is implemented at high level to indirectly control the contact force commanded by the user and track the target position in X^tY^t plane. Finally, the five body-frame force and torque inputs will be used to generate the propeller thrust force commands and the tilt angle set points to be fed into the rotors and servos. It is noted that since the system has 5-DOF independent actuation (one more DOF than traditional fix-rotor quadcopters), only the desired pitch angle θ_d is indirectly determined inside the control loop, while the desired setpoints for all the other DOFs are either determined from the target position/orientation or specified by the user. Specifically, $x_d^t = y_d^t = 0$ to let the UAV follow the target position in X^tY^t plane, z_d^t comes from the user-specified contact force setpoint to be discussed later in the impedance control design, the desired roll angle ϕ_d is specified by the user, and $\psi_d = \angle(proj(Z^t, X^eY^e), X^e)$ guarantees the manipulator side of the vehicle is facing the target.

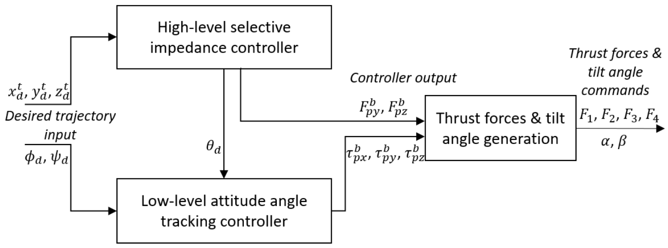


Fig. 2. The schematics of the proposed dual-level control strategy.

A. Low-level Attitude Control

The adaptive robust control (ARC) has demonstrated its powerfulness in many industrial applications that require precision motion control [18]–[20]. Specifically, ARC can effectively deal with both parametric uncertainties and uncertain nonlinearities in the system and accurate motion tracking. Thus, ARC is a perfect choice of controller for the low-level attitude angle control of the drilling/screwing operation in which a high tracking accuracy is required even in the presence of strong contact force/torque uncertainties. To proceed with the proposed ARC design, we first define

$$\begin{array}{rcl} \Theta & = & [d_{\tau x}^b,\ d_{\tau y}^b,\ d_{\tau z}^b]^T, \\ \tilde{\tau}_{\Delta}^b & = & \tau_{\Delta}^b - d_{\tau}^b = \tau_{\Delta}^b - [d_{\tau x}^b,\ d_{\tau y}^b,\ d_{\tau z}^b]^T, \end{array} \tag{7}$$

where Θ is the vector of constant portion of the lumped torque uncertainties to be adapted, and $\tilde{\tau}_{\Delta}^b$ is the time-varying portion. The following practical assumption is made on the uncertainties [18]:

Assumption 1: The extent of the parametric uncertainties and uncertain nonlinearities are known, i.e.,

$$\begin{array}{c} \Theta \in \Omega_{\Theta} \triangleq \{\Theta : \Theta_{min} \leq \Theta_L \leq \Theta_{max}\}, \\ \tilde{\tau}_{\Delta}^b \in \Omega_{\tilde{\tau}_{\Delta}^b} \triangleq \{\tilde{\tau}_{\Delta}^b : |\tilde{\tau}_{\Delta}^b| \leq \delta_{\tau}\}, \end{array} \tag{8}$$

where $\Theta_{min} = [\Theta_{1min}, \Theta_{2min}, \Theta_{3min}]^T$, $\Theta_{max} = [\Theta_{1max}, \Theta_{2max}, \Theta_{3max}]^T$, and δ_{τ} are the known bounds of the uncertainties.

Let $\hat{\Theta}$ denote the estimate of Θ, and $\tilde{\Theta}$ the estimation error (i.e., $\tilde{\Theta} = \hat{\Theta} - \Theta$). The following adaptation law with discontinuous projection modification is used:

$$\dot{\hat{\Theta}} = Proj_{\hat{\Theta}}(\Gamma \sigma), \tag{9}$$

where $\Gamma > 0$ is the diagonal matrix of adaptation gains, σ is the adaptation function to be synthesized later. The projection mapping $Proj_{\hat{*}}(\bullet) = [\cdots \quad Proj_{\hat{*}_i}(\bullet_i) \quad \cdots]^T$ is defined as

$$Proj_{\hat{*}_i}(\bullet_i) = \begin{cases} 0, & \text{if } \hat{*}_i = *_{imax} \text{ and } \bullet_i > 0, \\ 0, & \text{if } \hat{*}_i = *_{imin} \text{ and } \bullet_i < 0, \\ \bullet_i, & \text{otherwise.} \end{cases} \tag{10}$$

It can be shown that for any adaptation functions σ_L and σ_H, the projection mappings used in (10) guarantees

$$\begin{array}{ll} \mathbf{P1} & \hat{\Theta} \in \Omega_{\Theta}, \\ \mathbf{P2} & \tilde{\Theta}^T(\Gamma_L^{-1} Proj_{\hat{\Theta}}(\Gamma \sigma) - \sigma) \leq 0. \end{array} \tag{11}$$

Let $\eta_d(t) = [\phi_d(t)\ \theta_d(t)\ \psi_d(t)]^T$ be the desired attitude angle trajectory, and $e_{\eta} = \eta - \eta_d$ be the attitude tracking error. We define the following switching-function-like quantity for the attitude tracking error as

$$s_{\eta} = \Psi^{-1}(\dot{e}_{\eta} + k_{\eta 1} e_{\eta}) = \omega^b - \Psi^{-1}\dot{\eta}_d + \Psi^{-1} k_{\eta 1} e_{\eta}, \tag{12}$$

where $k_{\eta 1} > 0$ is a diagonal gain matrix. Clearly, if s_{η} is small or converges to zero exponentially, then the attitude tracking error e_{η} will be small or converge to zero exponentially. So the rest of the design is to make $s_{\eta} \to 0$. Differentiating (12) and using (5):

$$\begin{array}{rcl} I^b \dot{s}_{\eta} & = & I^b \dot{\omega}^b - I^b[\dot{\Psi}^{-1}\dot{\eta}_d + \Psi^{-1}\ddot{\eta}_d - \dot{\Psi}^{-1} k_{\eta 1} e_{\eta} \\ & & -\Psi^{-1} k_{\eta 1}(\Psi \omega^b - \dot{\eta}_d)] \\ & = & \tau_p^b + \Theta + B + \tilde{\tau}_{\Delta}^b, \end{array} \tag{13}$$

where $B = -(I^b\omega^b) \times \omega^b - I^b[\dot{\Psi}^{-1}\dot{\eta}_d + \Psi^{-1}\ddot{\eta}_d - \dot{\Psi}^{-1}k_{\eta 1}e_{\eta} - \Psi^{-1}k_{\eta 1}(\Psi\omega^b - \dot{\eta}_d)]$ is the known nonlinear term to be compensated directly.

The following control law is proposed:

$$\tau_p^b = \tau_{ps}^b + \tau_{pa}^b, \quad \tau_{pa}^b = -\hat{\Theta} - B, \tag{14}$$

where τ_{pa}^b is the adjustable disturbance compensation term, τ_{ps}^b is the robust feedback term synthesized as

$$\tau_{ps}^b = \tau_{ps1}^b + \tau_{ps2}^b, \quad \tau_{ps1}^b = -k_{\eta 2} s_{\eta}, \tag{15}$$

where τ_{ps1}^b is a proportional feedback term with $k_{\eta 2} > 0$ being the diagonal gain matrix. τ_{ps2}^b is a robust feedback to attenuate the effect of nonlinear model uncertainties caused by parameter estimation error, which satisfies the following two conditions:

$$\begin{array}{ll} 1. & s_{\eta}^T(\tau_{ps2}^b - \tilde{\Theta}_L + \tilde{\tau}_{\Delta}^b) \leq \epsilon_{\eta}, \\ 2. & s_{\eta}^T \tau_{ps2}^b \leq 0, \end{array} \tag{16}$$

where ϵ_{η} is a design parameter which can be arbitrarily small and τ_{ps2} can be chosen according to the previous research [18]. Similar to [18], theoretical results on the robust performance and asymptotic tracking can be obtained, which are omitted here due to page limit.

B. High-level Selective Impedance Control

In the high level, the 3-DOF translation of the vehicle's body is controlled such that the end-effector applies the user-specified desired contact force F_{cd} normal to the workpiece surface, while the coordinates along the two other DOFs closely track the point to drill or screw in. To achieve this goal, we first transform the position coordinates of the vehicle from inertia frame to the target frame whose origin coincides with the point to drill/screw and Z^t axis is perpendicular to the workpiece surface, then derive the decoupled system dynamics along X^t, Y^t, Z^t directions of the target frame. Based on this dynamics, a selective impedance control is designed such that the user can specify different impedance parameters to achieve different force-position characteristics along different axes of the target frame.

To begin with, let us define $\lambda^t = [x^t\ y^t\ z^t]^T$ as the coordinate vector of vehicle's position O_b expressed in the target frame. A coordinate transformation leads to the following relationship:

$$\lambda^t = R_e^t(\lambda - \lambda_t), \tag{17}$$

where R_e^t is the rotation matrix from target frame to inertia frame. λ_t is the coordinate of the target frame origin O_t expressed in inertia frame. Both R_e^t and λ_t are constants for a particular drilling/screwing operation. Differentiating the above equation twice, and applying (5), (6), we have

$$M\ddot{\lambda}^t = F_p^t + F_g^t + F_c^t, \tag{18}$$

where F_p^t, F_g^t, F_c^t are the total thrust force, gravity force, and contact force represented in target frame, respectively.

The selective impedance controller is designed as

$$F_p^t = -F_g^t - F_c^t + M\ddot{\lambda}_d^t - C\dot{e}_\lambda^t - Ke_\lambda^t, \tag{19}$$

where $e_\lambda^t = \lambda^t - \lambda_d^t$ is the position tracking error in target frame, λ_d^t is the desired position, $C = diag(C_x, C_y, C_z)$ and $K = diag(K_x, K_y, K_z)$ are diagonal matrices consisting of all the virtual damping and spring parameters along X^t, Y^t, Z^t. With the proposed design, it is clear that the following impedance model relating the position tracking error to the contact force holds true:

$$M\ddot{e}_\lambda^t + C\dot{e}_\lambda^t + Ke_\lambda^t = F_c^t. \tag{20}$$

The damping and spring parameters for different axes can be selected differently to achieve the hybrid motion/force control objective. For the proposed drilling/screwing application, K_x and K_y are chosen to be big enough to ensure an accurate target point tracking on the tangential plane if λ_{dx}^t and λ_{dy}^t are both set as zero. Since the tangential contact force is normally very small, the position tracking error $e_{\lambda x}^t$ and $e_{\lambda y}^t$ will exponentially converge to zero. In comparison, K_z is selected to be relatively small to make the impedance model more flexible along the normal direction, allowing an efficient indirect normal contact force regulation by properly tuning the desired position. It can be seen that at steady state, the contact force along Z^t converges to its desired setpoint F_d^t if the desired position along Z^t is set to be $z_d^t = z^t - \frac{F_d^t}{K_z}$. After all the spring constants for the impedance model are set, the damping constants will be selected to maximize the transient performance. For example, C can be chosen as $1.414\sqrt{KM}$ so that the closed-loop damping ratio of each axis is 0.707, which gives the maximum bandwidth without any resonant peak in the bode magnitude plot.

After synthesizing the target frame thrust force F_p^t, the inertia frame thrust force can be calculated as $F_p^e = R_t^e F_p^t$. To achieve this required thrust force in inertia frame, the body-frame thrust components F_{py}^b, F_{pz}^b as well as the desired pitch angle θ_d fed into the low-level controller are calculated as follows

$$\begin{aligned}
F_{py}^b &= (F_{px}^e C_\psi + F_{py}^e S_\psi)^2 S_\phi + {F_{pz}^e}^2 S_\phi \\
&\quad + (F_{px}^e S_\psi - F_{pz}^e C_\psi) C_\phi, \\
F_{pz}^b &= (F_{px}^e C_\psi + F_{py}^e S_\psi)^2 C_\phi + {F_{pz}^e}^2 C_\phi \\
&\quad + (F_{px}^e S_\psi - F_{py}^e C_\psi) C_\phi, \\
\theta_d &= atan\left(\frac{F_{px}^e C_\psi + F_{py}^e S_\psi}{F_{pz}^e}\right).
\end{aligned} \tag{21}$$

Finally, by setting $\alpha = \beta$ and solving the five equations in (2) and (4), the individual rotor thrust forces and the tilt angle can be obtained as:

$$\begin{aligned}
&\alpha = \beta = atan(\tfrac{F_{py}^b}{F_{pz}^b}) \\
&F_1 = \frac{F_{pz}^b k l_1 l_2 + k l_1 \tau_{px}^b + {C_a}^2 l_2 (k\tau_{py}^b + l_1 \tau_{pz}^b) + C_a S_a l_2 (l_1 \tau_{py}^b - k\tau_{pz}^b)}{4C_a k l_1 l_2} \\
&F_2 = \frac{F_{pz}^b k l_1 l_2 - k l_1 \tau_{px}^b + {C_a}^2 l_2 (k\tau_{py}^b - l_1 \tau_{pz}^b) - C_a S_a l_2 (l_1 \tau_{py}^b + k\tau_{pz}^b)}{4C_a k l_1 l_2} \\
&F_3 = \frac{F_{pz}^b k l_1 l_2 - k l_1 \tau_{px}^b - {C_a}^2 l_2 (k\tau_{py}^b - l_1 \tau_{pz}^b) + C_a S_a l_2 (l_1 \tau_{py}^b + k\tau_{pz}^b)}{4C_a k l_1 l_2} \\
&F_4 = \frac{F_{pz}^b k l_1 l_2 + k l_1 \tau_{px}^b - {C_a}^2 l_2 (k\tau_{py}^b + l_1 \tau_{pz}^b) - C_a S_a l_2 (l_1 \tau_{py}^b - k\tau_{pz}^b)}{4C_a k l_1 l_2}.
\end{aligned} \tag{22}$$

V. Vision-based target identification and tracking

During the aerial drilling/screwing operation, the end-effector needs to be aligned with the target point. This process could be extremely time consuming and inaccurate if conducted by a human operator who takes action based on the visual feedback information sent from the onboard camera. To overcome this challenge, a real-time vision-based target identification and tracking method is proposed, and is integrated with our human-guided manipulation to help improve positioning accuracy and efficiency for drilling and screwing. Under the proposed control framework, the vision-based target identification and tracking system aims at obtaining the relative position and orientation of the target frame w.r.t camera frame (λ_t^c and R_t^c) first, and subsequently the tracking error e_λ^t to be used in the high-level impedance controller. The vision system consists of the following three steps.

A. Target Identification Using a YOLO v3 Object Detector

The YOLO v3 detector eliminates the RoI extraction stage, and adopts a unified architecture that directly extracts feature maps from input images, then it regards the whole feature maps as candidate regions to predict bounding boxes and categories [21], [22]. Its characteristics of realtime object identification and high detection precision, make it a perfect solution for our object detection task in unstructured environments. Additionally, the target workpiece to be operated in real-world conditions may be in different sizes or trimmed to fit different environments, a more general and robust detector can be trained with more labeled images taken from different applications and environments. Each detector model utilized in this study is trained before the task based on a set of 600 images of the target taken from different distances and angles, and labeled by ourselves. This object detector will search for the target in the view of camera and predict a bounding box to indicate the position of the target in the image.

B. Target Tracking Using a Kanade-Lucas-Tomasi (KLT) Tracker

Although YOLO is a robust and powerful object detector, the outcoming bounding box does not estimate the rotation of object caused by the change of UAV's pitch angle θ. Besides, the DNN-based YOLO object detector is computationally expensive since it extracts feature maps from the entire input image. Thus, to track the target position and orientation in real time, we use the computationally efficient Kanade-Lucas-Tomasi (KLT) algorithm that can estimate and track all basic 2D transformations of the target in a small local region of the input image [23].

Following the first step, once the target workpiece is located by YOLO v3 object detector, a bounding box is inserted around the workpiece to indicate a small template image for tracking, the next step is to extract feature points within the bounding box that can be reliably tracked, using the proposed method from [24]. The type of 2D transformation to be tracked in this work is called similarity including translation, rotation and scaling, which can be expressed as

$$\boldsymbol{x}' = \begin{bmatrix} s\boldsymbol{R} & \boldsymbol{t} \end{bmatrix} \overline{\boldsymbol{x}} \tag{23}$$

where $\boldsymbol{x}'$ is the point coordinate in image plane after transformation, $\overline{\boldsymbol{x}}$ is the homogenous coordinate before transformation, s is an arbitrary scale factor, $\boldsymbol{R}$ is an orthonormal rotation matrix, $\boldsymbol{t} = [t_x, t_y]^T$ is the translation vector.

To better parameterize the transformation for tracking, the warping function is defined as

$$\mathbf{W}(\mathbf{x};\mathbf{p}) = \begin{bmatrix} a & -b & t_x \\ b & a & t_y \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \tag{24}$$

where $\mathbf{x} = [x, y]^T$, and $\mathbf{p} = \begin{bmatrix} t_x, t_y, a, b \end{bmatrix}^{\mathrm{T}}$ is the vector of transformation parameters. At every time instance, the optimal parameter increase $\Delta\mathbf{p}$ is calculated by solving the following the problem minimizing the feature point difference between the template image and warped-back image [25]:

$$\min_{\Delta\mathbf{p}} \sum_{\mathbf{x}} [I(\mathbf{W}(\mathbf{x};\mathbf{p}+\Delta\mathbf{p})) - T(\mathbf{x})]^2, \tag{25}$$

where $T(\mathbf{x})$ is template image. $I(\mathbf{W}(\mathbf{x};\mathbf{p}+\Delta\mathbf{p}))$ is input image warped back onto the coordinate frame of the template.

The parameters $\mathbf{p}$ will be updated iteratively according to $\Delta\mathbf{p}$, and transformation is then applied to the bounding box around the target feature for real-time tracking.

C. Morphological Image Processing to Obtain the Target Coordinate

The objective in the final step is to acquire the camera-frame coordinate of the point of interest for drilling/screwing on the target feature. This goal can be achieved by investigating the structure of the feature and extracting geometrical elements from the structure. However, numerous noises and errors would be introduced into the image by disturbances from the unstructured environments. Thus, the morphological image processing method [26] is applied to remove the noises and enhance image quality by accounting for the form and structure of the image, furthermore, to generate accurate coordinate information of the point of interest in camera frame.

Customized morphological structuring elements can be defined to locate the specific geometry element and the point for drilling and screwing on the target feature. After that, the position of target point $\begin{bmatrix} x_t^c & y_t^c & z_t^c \end{bmatrix}^{\mathrm{T}}$ expressed in camera

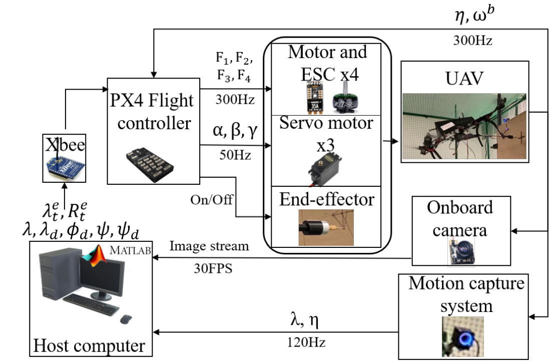


Fig. 3. The overall system architecture of the experiment, showing the key components and the data flow.

frame can be derived from the pixel coordinate(u_i, v_i) of it in image plane through the follow relation:

$$\begin{bmatrix} x_t^c \\ y_t^c \\ z_t^c \end{bmatrix} = z_t^c \begin{bmatrix} \alpha_x f & sf & p_x \\ 0 & \alpha_y f & p_y \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} u_{\mathrm{i}} \\ v_{\mathrm{i}} \\ 1 \end{bmatrix}, \tag{26}$$

where f is the focal length, α_x, α_y are pixel scaling factors, (p_x, p_y) is the principle point (where optical axis hits image plane) , s is the slant factor when the image plane is not normal to the optical axis.

VI. Experiments

In this section, the proposed novel UAV platform for aerial drilling and screwing are experimentally validated at the Assistive and Intelligent Robotics Lab of NJIT. Experimental setup is first illustrated, then followed by experimental results and discussions. The video of the experiments is available at https://youtu.be/uw6Aw78jlgY.

A. Experimental setup

The newly designed aerial manipulator used in the experiment is equipped with an ARM Cortex based PX4 Pixhawk2 flight controller for computing the control input and communicating with ground host computer through an XBee module, as shown in Fig. 3. To detect the target object, a zero-latency all-in-one camera (including an FPV camera, a transmitter and an antenna) is installed with eye-in-hand configuration. The camera has about $120°$ FOV and the resolution of the image is 640×480 with 30 fps. An OptiTrack motion capture system is set up around the flight testing area to acquire attitude and position measurement of the vehicle at $120Hz$, and send it to the host computer. The host computer also receives video from the onboard camera in realtime through a $5.8G$ $150CH$ UVC receiver. Based on received video and motion capture information, the proposed vision-based identification and tracking algorithm is running on the host computer, and commands are sent to the UAV through Xbee.

B. Experimental results and discussions

First, to verify the vehicle's decoupled motion and force control performances on the longitudinal plane, which are

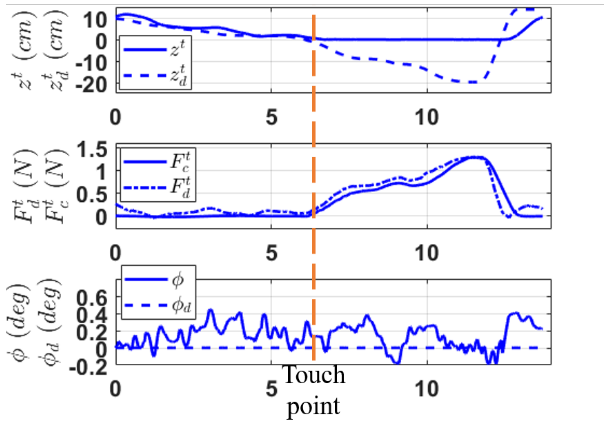
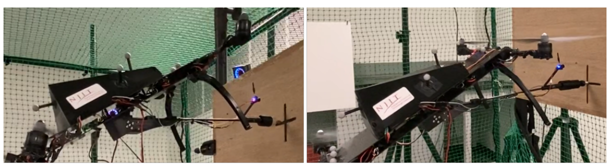


Fig. 4. Experimental characterizations of the impedance controller and the attitude controller. Touch point is where the arm begins touching the wood board.



(a) Drilling (b) Screwing

Fig. 5. Aerial hole drilling and screwing with the proposed human-guided semi-automated aerial drilling/screwing platform

regulated by the low-level ARC attitude control and the high-level selective impedance control respectively, we carry out a contact force test by pushing the manipulator arm (with end-effector removed) against a wood board attached to a 6-axis ATI F/T sensor while maintaining a stable body hovering. As shown in Fig. 4, before the end-effector touches the board, the forward motion is well controlled with z^t close to z_d^t. After contact is established, z^t stays constant, while $z_d^t = z^t - \frac{F_d^t}{K_z}$ penetrates into the board to generate the required contact force. The corresponding force command $F_d^t = K_z(z^t - z_d^t)$ matches the real measured force F_c^t very well. The independent rotational motion (ϕ) keeps stationary during the whole test, even as the manipulator arm is in contact with the board, showing the reliability of the controller.

Next, we conduct three tests for both the aerial hole drilling and the bolt screwing tasks, as shown in Fig. 5 and illustrated in the video file attached. A 12mm thick wood board is used as the workpiece and is placed vertically in the lab zone. For demonstration purpose, we simply draw cross-shaped center marks on the board to represent the target locations for drilling/screwing operations. After the aerial manipulator is guided close to the board, and a proper view of the cross marker is obtained, the vision-based identification and tracking system is activated and successfully implemented, as shown in Fig. 6. The position tracking error between the target point and the end-effector, captured by the vision guidance system, expressed in the target frame is shown in Fig. 7. The plots demonstrate that the vision-based guidance system can successfully capture the position of the target and

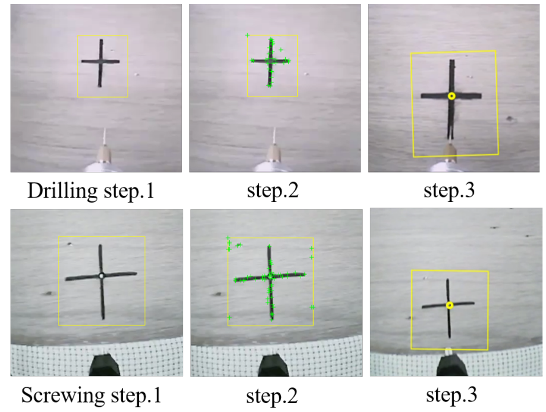


Fig. 6. The three steps of vision-based positioning and tracking method: the first step is target object detection by a YOLO v3 detector, the second step is feature points extraction and 2D transformation tracking using KLT tracker, the third step is to capture the point of interest for drilling/screwing.

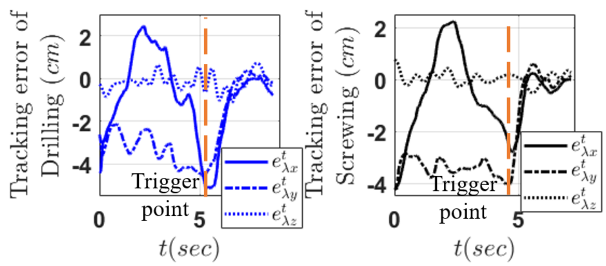


Fig. 7. Position of the target point with respect to end-effector estimated by the vision guidance system and expressed in the target frame. At the trigger point, command is sent to align the end-effector tool with the target point.

output smooth tracking error, enabling accurate and fast target tracking for drilling/screwing.

After the drill/screwdriver is aligned with the target point, a human operator guides the aerial manipulator to conduct feeding operation by tuning the contact force command. The selective impedance control law running on host computer takes the contact force command as input and automatically generates desired position commands to indirectly regulate the contact force between the end-effector and wood board along the feeding direction. As can be seen in Fig. 8, to approach to the wood board, very light and short-time force commands are given to gradually drive the aerial manipulator forward. After contact is established, strong and continuous contact force is exerted to execute and accelerate the drilling/screwing operation. It can be found that the drilling requires a stronger contact force to penetrate wood board surface compared with the screwing process. Overall, the experimental results show the effectiveness of the proposed platform with design, control and sensing in aerial hole drilling and bolt screwing. The accuracy over all the three drilling tests is within $\pm 2.5mm$, as shown in Fig. 9.

VII. Conclusion

This paper presented the design, sensing, and control of a novel UAV platform for omnidirectional aerial drilling and screwing. To solve the coupling between translational and

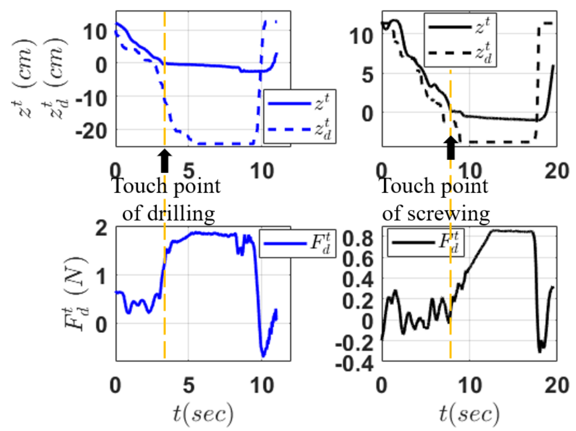


Fig. 8. The force command F_d^t along Z^t for drilling (left) and screwing (right). Touch point is where the end-effector begins touching the wood board.

Fig. 9. The drilling accuracy over all the three aerial drilling tests.

rotational motions of aerial manipulators based on traditional multirotors, a novel 5-DOF actuated energy efficient aerial manipulator was designed with the capability of independently exerting strong drilling/screwing force to the workpiece along any direction on the longitudinal plane. A dual-level control architecture was proposed, with a low-level adaptive robust control (ARC) to accurately track the attitude angles in the presence of force/torque uncertainties, and a high level selective impedance control to indirectly control the contact force commanded by the user and track the target position. A vision-based identification and tracking scheme was developed by integrating a robust YOLO v3 object detector with feature tracking and morphological techniques, achieving automatically alignment between the drill/screwdriver and target point. Experimental results conducted indoor under the motion capture system were presented to verify the proposed aerial manipulation platform. In the future, we expect to use onboard sensors for localization and apply the platform to various challenging outdoor applications in construction, decoration, transportation, or maintenance at hard-to-reach locations.

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